Exercises

Exercise 1 : SQL Fundamentals

Database : employees_db

ID	firstName	lastName	department	Salary	hire-date	City
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	58000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

Questions

1. SELECT Statement

Write an SQL query to retrieve all colums from employees table

SQL

```
SELECT *
FROM employees;
```

2. SELECT DISTINCT

SQL

```
SELECT DISTINCT department
  FROM employees;
```

3 ORDER BY statement

SQL

```
SELECT first_name,
       last_name
FROM employees
  ORDER BY salary DESC;
```

4. LIMIT STATEMENT

SQL

```
SELECT * FROM employees
  ORDER BY salary DESC
    LIMIT 5;
```

5) WHERE STATEMENT

SQL

```
SELECT *
  FROM employees
  WHERE department = 'IT';
```

6 AND statement

SQL

```
SELECT * FROM employees
WHERE department = 'Finance' AND salary > 5~~6~~000 55000
```

7 OR STATEMENT

SQL

```
SELECT *
 FROM employees
WHERE department = 'HR' OR department = 'Marketing';
```

8 NOT Statement

SQL

```
SELECT *
 FROM employees
 WHERE NOT department = 'IT';
```

9 IN STATEMENT

SQL

```
SELECT *
 FROM employees
 WHERE department IN ('HR', 'IT', 'Finance');
```

10 Combining Conditions ~~WHERE, AND, ORDER BY~~

SQL

```
SELECT *
 FROM employees
 WHERE department = 'IT'
  AND salary > 50 000
  AND city = 'New_York';
```

11 Combining WHERE, AND and ORDER BY

SQL

```
SELECT first_name,
       last_name
FROM employees
WHERE (department = 'Finance' OR department = 'Marketing')
  AND Salary > 52000
ORDER BY salary DESC;
```

12 Combining SELECT DISTINCT, WHERE IN

SQL

```
SELECT DISTINCT city
FROM employees
WHERE department NOT IN ~~('IT')~~ ('IT', 'HR');
```

13 Combining WHERE, ~~OR, IN and LIMIT~~

```
SELECT *
FROM employees
~~WHERE department NOT~~
WHERE NOT department = 'Finance'
  AND salary > 50000
ORDER BY hire_date ASC;
```

14 Combining WHERE, OR, IN and LIMIT

13. Combining WHERE, ~~OR, IN~~ NOT, AND and ORDER BY

SQL

```
SELECT * FROM employees
 WHERE NOT department = 'Finance'
  AND salary > 50000
   ORDER BY hire-date ASC;
```

14 Combining WHERE, OR, IN, and LIMIT

SQL

```
 SELECT *
 FROM employees
WHERE city IN ('chicago', 'Los Angeles')
  AND department IN ('IT, 'Marketing')
    LIMIT 3;
```

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